

Embeddings & Semantic Search — Research Report

Task 2 — Week 2, Resume AI Project

1. Embeddings

An embedding is a way of turning text (a word, sentence, or paragraph) into a list of numbers, called a vector. These numbers are not random — they are generated so that pieces of text with similar meaning end up with similar numbers. For example, the embedding for 'dog' and the embedding for 'puppy' will be numerically close to each other, while the embedding for 'dog' and 'airplane' will be far apart.

In short, embeddings let a computer represent meaning as points in space. Two ideas with similar meaning sit close together in that space, even if the actual words used are completely different.

2. Sentence Transformers

A Sentence Transformer is a type of AI model specifically trained to convert whole sentences (not just single words) into embeddings. Earlier embedding methods worked mainly at the word level, which made it hard to capture the meaning of a full sentence. Sentence Transformers solve this by producing one embedding that represents the meaning of an entire sentence or paragraph.

This is especially useful for this project, because it allows the system to compare a full resume against a full job description as single units of meaning, rather than comparing word by word.

3. Vector Databases

Once text has been converted into embeddings (vectors), those vectors need to be stored somewhere so they can be searched later. A vector database is a specialized type of database built specifically to store embeddings and quickly find the ones that are most similar to a given query.

Instead of searching by exact text match like a traditional database, a vector database searches by numerical closeness. This makes it possible to ask 'find me resumes similar in meaning to this job description' and get back accurate, ranked results very quickly, even across thousands of documents.

4. Cosine Similarity

Cosine similarity is the mathematical method most commonly used to measure how close two embeddings are to each other. It looks at the angle between two vectors rather than their exact position, and produces a score, usually between -1 and 1 (or shown as a percentage).

- A score close to 1 (or 100%) means the two pieces of text have very similar meaning.
- A score close to 0 means the two pieces of text are unrelated.
- A score close to -1 means the meanings are close to opposite (rare in most text comparisons).

This is the exact method the project's similarity engine (Task 4) will use to compare a resume embedding against a job description embedding, producing the final match percentage.

5. Semantic Search

Semantic search is a search method that looks for meaning rather than exact words. A traditional search engine might only return results that contain the exact word typed in. A semantic search engine, powered by embeddings, can understand the intent behind a query and return results that are related in meaning, even if they use different words entirely.

For example, searching 'affordable laptop for students' with semantic search could return results mentioning 'budget-friendly notebook for college use', because the system understands these phrases mean nearly the same thing.

6. Why Keyword Matching is Insufficient

Keyword matching only checks for exact word overlap. This creates several real problems:

- Synonyms are missed — 'teamwork' and 'collaboration' mean the same thing, but keyword matching may treat them as unrelated.
- Different phrasing is penalized — a candidate who writes 'led a team of 5' instead of 'team leadership' may be scored lower, despite having the exact skill.
- It rewards keyword stuffing — candidates can artificially boost their score by repeating exact words from the job posting, even if they don't truly have that experience.
- It cannot understand context — the word 'python' could mean the programming language or the snake, and keyword matching alone cannot tell the difference.

Because of these issues, relying only on keyword matching can unfairly reject qualified candidates and let poorly matched resumes rank higher just because of clever wording. This is the core reason this project uses embeddings and semantic similarity instead of simple keyword counting.

7. How Embeddings Represent Meaning

Embeddings are created by AI models that have been trained on huge amounts of text. During training, the model learns patterns in how words and sentences are used together. Over time, it learns to place text with similar meaning close together in a multi-dimensional numerical space, and text with different meaning far apart.

Each embedding is typically a list of a few hundred numbers (for example, 384 or 768 numbers, depending on the model). While individual numbers don't mean anything on their own, taken together, the whole list captures the meaning of the text in a form a computer can compare mathematically.

8. Real-World Applications of Semantic Search

- Search engines — modern search engines use semantic search to return relevant results even when the exact words don't match the query.
- Customer support chatbots — matching a user's question to the most relevant help article, even if worded differently.
- E-commerce — recommending products based on the meaning of a search, not just exact product names.
- Recruiting and hiring — matching resumes to job descriptions based on actual skill and experience overlap (this project's use case).
- Document and legal search — finding relevant contracts, case law, or research papers based on topic, not just exact terms.
- Plagiarism and duplicate detection — identifying content that means the same thing even if reworded.